Neural Computation -- 2025

62 article(s), all issues

1. Closed-Loop Multistep Planning

Giulia Lafratta, Bernd Porr, Christopher Chandler, Alice Miller

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Living organisms interact with their surroundings in a closed-loop fashion, where sensory inputs dictate the initiation and termination of behaviors. Even simple animals are able to develop and execute complex plans, which has not yet been replicated in robotics using pure closed-loop input control. We propose a solution to this problem by defining a set of discrete and temporary closed-loop controllers, called "Tasks," each representing a closed-loop behavior. We further introduce a supervisory module that has an innate understanding of physics and causality, through which it can simulate the execution of Task sequences over time and store the results in a model of the environment. On the basis of this model, plans can be made by chaining temporary closed-loop controllers. Our proposed framework was implemented for a robot and tested in two scenarios as proof of concept.

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2. Neural Associative Skill Memories for Safer Robotics and Modeling Human Sensorimotor Repertoires

Pranav Mahajan, Mufeng Tang, Tianjin Li, Ioannis Havoutis, Ben Seymour

https://doi.org/10.1162/neco.a.1475 | pp. 1-27

Modern robots face a challenge shared by biological systems: how to learn and adaptively express multiple sensorimotor skills. A key aspect of this is developing an internal model of expected sensorimotor experiences to detect and react to unexpected events, guiding self-preserving behaviors. Associative skill memories (ASMs) address this by linking movement primitives to sensory feedback, but existing implementations rely on hard-coded libraries of individual skills. A key unresolved problem is how a single neural network can learn a repertoire of skills while enabling integrated fault detection and context-aware execution. Here we introduce neural associative skill memories (neural ASMs), a framework that uses self-supervised temporal predictive coding to integrate skill learning and expression using biologically plausible local learning rules. Unlike traditional ASMs, which require explicit skill selection, neural ASMs implicitly recognize and express skills through contextual inference, enabling fault detection using "predictive surprise" across the entire learned repertoire. Compared to recurrent neural networks trained using backpropagation through time, our model achieves comparable qualitative performance in skill memory expression while using local learning rules and predicts a biologically relevant speed-versus-accuracy trade-off. By integrating fault detection, reactive control, and skill expression into a single energy-based architecture, neural ASMs contribute to safer, self-preserving robotics and provide a computational lens to study biological sensorimotor learning.

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3. Working Memory and Self-Directed Inner Speech Enhance Multitask Generalization in Active Inference

Jeffrey Queißer, Jun Tani

https://doi.org/10.1162/neco.a.36 | pp. 28-70

This simulation study shows how a set of working memory tasks can be acquired simultaneously through interaction between a stacked recurrent neural network (RNN) and multiple working memories. In these tasks, temporal patterns are provided, followed by linguistically specified task goals. Training is performed in a supervised manner by minimizing the free energy, and goal-directed tasks are performed using the active inference (AIF) framework. Our simulation results show that the best task performance is obtained when two working memory modules are used instead of one or none and when self-directed inner speech is incorporated during task execution. Detailed analysis indicates that a temporal hierarchy develops in the stacked RNN module under these optimal conditions. We argue that the model's capacity for generalization across novel task configurations is supported by the structured interplay between working memory and the generation of self-directed language outputs during task execution. This interplay promotes internal representations that reflect task structure, which in turn support generalization by enabling a functional separation between content encoding and control dynamics within the memory architecture.

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4. Effective Learning Rules as Natural Gradient Descent

Lucas Shoji, Kenta Suzuki, Leo Kozachkov

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We establish that a broad class of effective learning rules-those that improve a scalar performance measure over a given time window-can be expressed as natural gradient descent with respect to an appropriately defined metric. Specifically, parameter updates in this class can always be written as the product of a symmetric positive-definite matrix and the negative gradient of a loss function encoding the task. Given the high level of generality, our findings formally support the idea that the gradient is a fundamental object underlying all learning processes. Our results are valid across a wide range of common settings, including continuous- time, discrete-time, stochastic, and higher-order learning rules, as well as loss functions with explicit time dependence. Beyond providing a unified framework for learning, our results also have practical implications for control as well as experimental neuroscience.

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5. Possible Principles for Aligned Structure Learning Agents

Lancelot Da Costa, Tomáš Gaven■iak, David C. Hyland, Mandana Samiei, Cristian Dragos-Manta, Candice Pattisapu, Adeel Razi, Karl Friston

https://doi.org/10.1162/neco.a.39 | pp. 97-143

This paper offers a road map for the development of scalable aligned artificial intelligence (AI) from first principle descriptions of natural intelligence. In brief, a possible path toward scalable aligned AI rests on enabling artificial agents to learn a good model of the world that includes a good model of our preferences. For this, the main objective is creating agents that learn to represent the world and other agents' world models, a problem that falls under structure learning (also known as causal representation learning or model discovery). We expose the structure learning and alignment problems with this goal in mind, as well as principles to guide us forward, synthesizing various ideas across mathematics, statistics, and cognitive science. We discuss the essential role of core knowledge, information geometry, and model reduction in structure learning and suggest core structural modules to learn a wide range of naturalistic worlds. We then outline a way toward aligned agents through structure learning and theory of mind. As an illustrative

example, we mathematically sketch Asimov's laws of robotics, which prescribe agents to act cautiously to minimize the ill-being of other agents. We supplement this example by proposing refined approaches to alignment. These observations may guide the development of artificial intelligence in helping to scale existing, or design new, aligned structure learning systems.

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6. Simulated Complex Cells Contribute to Object Recognition Through Representational Untangling

Mitchell B. Slapik, Harel Z. Shouval

https://doi.org/10.1162/neco.a.1480 | pp. 145-164

The visual system performs a remarkable feat: it takes complex retinal activation patterns and decodes them for object recognition. This operation, termed "representational untangling," organizes neural representations by clustering similar objects together while separating different categories of objects. While representational untangling is usually associated with higher-order visual areas like the inferior temporal cortex, it remains unclear how the early visual system contributes to this process-whether through highly selective neurons or high-dimensional population codes. This article investigates how a computational model of early vision contributes to representational untangling. Using a computational visual hierarchy and two different data sets consisting of numerals and objects, we demonstrate that simulated complex cells significantly contribute to representational untangling for object recognition. Our findings challenge prior theories by showing that untangling does not depend on skewed, sparse, or high-dimensional representations. Instead, simulated complex cells reformat visual information into a low-dimensional, yet more separable, neural code, striking a balance between representational untangling and computational efficiency.

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7. Approximation Rates in Fréchet Metrics: Barron Spaces, Paley-Wiener Spaces, and Fourier Multipliers

Ahmed Abdeljawad, Thomas Dittrich

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Operator learning is a recent development in the simulation of partial differential equations by means of neural networks. The idea behind this approach is to learn the behavior of an operator, such that the resulting neural network is an approximate mapping in infinite-dimensional spaces that is capable of (approximately) simulating the solution operator governed by the partial differential equation. In our work, we study some general approximation capabilities for linear differential operators by approximating the corresponding symbol in the Fourier domain. Analogous to the structure of the class of Hörmander symbols, we consider the approximation with respect to a topology that is induced by a sequence of semi-norms. In that sense, we measure the approximation error in terms of a Fréchet metric, and our main result identifies sufficient conditions for achieving a predefined approximation error. We then focus on a natural extension of our main theorem, in which we reduce the assumptions on the sequence of seminorms. Based on existing approximation results for the exponential spectral Barron space, we then present a concrete example of symbols that can be approximated well.

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8. Unsupervised Learning in Echo State Networks for Input Reconstruction

Taiki Yamada, Yuichi Katori, Kantaro Fujiwara

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Echo state networks (ESNs) are a class of recurrent neural networks in which only the readout layer is trainable, while the recurrent and input layers are fixed. This architectural constraint enables computationally efficient processing of time-series data. Traditionally, the readout layer in ESNs is trained using supervised learning with target outputs. In this study, we focus on input reconstruction (IR), where the readout layer is trained to reconstruct the input time series fed into the ESN. We show that IR can be achieved through unsupervised learning (UL), without access to supervised targets, provided that the ESN parameters are known a priori and satisfy invertibility conditions. This formulation allows applications relying on IR, such as dynamical system replication and noise filtering, to be reformulated within the UL framework via straightforward integration with existing algorithms. Our results suggest that prior knowledge of ESN parameters can reduce reliance on supervision, thereby establishing a new principle-not only by fixing part of the network parameters but also by exploiting their specific values. Furthermore, our UL-based algorithms for input reconstruction and related tasks are suitable for autonomous processing, offering insights into how analogous computational mechanisms might operate in the brain in principle. These findings contribute to a deeper understanding of the mathematical foundations of ESNs and their relevance to models in computational neuroscience.

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9. Sum-of-Norms Regularized Nonnegative Matrix Factorization

Andersen Ang, Waqas Bin Hamed, Hans De Sterck

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When applying nonnegative matrix factorization (NMF), the rank parameter is generally unknown. This rank, called the nonnegative rank, is usually estimated heuristically since computing its exact value is NP-hard. In this work, we propose an approximation method to estimate the rank on the fly while solving NMF. We use the sum-of-norm (SON), a group-lasso structure that encourages pairwise similarity, to reduce the rank of a factor matrix when the initial rank is overestimated. On various data sets, SON-NMF can reveal the correct nonnegative rank of the data without prior knowledge or parameter tuning. SON-NMF is a nonconvex, nonsmooth, nonseparable, and nonproximable problem, making it nontrivial to solve. First, since rank estimation in NMF is NP-hard, the proposed approach does not benefit from lower computational complexity. Using a graph-theoretic argument, we prove that the complexity of SON NMF is essentially irreducible. Second, the per iteration cost of algorithms for SON-NMF can be high. This motivates us to propose a first-order BCD algorithm that approximately solves SON-NMF with low per iteration cost via the proximal average operator. SON-NMF exhibits favorable features for applications. Besides the ability to automatically estimate the rank from data, SON-NMF can handle rank-deficient data matrices and detect weak components with little energy. Furthermore, in hyperspectral imaging, SON-NMF naturally addresses the issue of spectral variability.

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10. Replay as a Basis for Backpropagation Through Time in the Brain

Huzi Cheng, Joshua W. Brown

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How episodic memories are formed in the brain is a continuing puzzle for the neuroscience community. The brain areas that are critical for episodic learning (e.g., the hippocampus) are characterized by recurrent connectivity and generate frequent offline replay events. The function of the replay events is a subject of active debate. Recurrent connectivity, computational simulations show, enables sequence learning when combined with a suitable learning algorithm such as backpropagation through time (BPTT). BPTT, however, is not biologically plausible. We describe here, for the first time, a biologically plausible variant of BPTT in a reversible recurrent neural network, R2N2, that critically leverages offline replay to support episodic learning. The model uses forward and backward offline replay to transfer information between two recurrent neural networks, a cache and a consolidator, that perform rapid one-shot learning and statistical learning, respectively. Unlike replay in standard BPTT, this architecture requires no artificial external memory store. This approach outperforms existing solutions like random feedback local online learning and reservoir network. It also accounts for the functional significance of hippocampal replay events. We demonstrate the R2N2 network properties using benchmark tests from computer science and simulate the rodent delayed alternation T-maze task.

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11. Improving Recall in Sparse Associative Memories That Use Neurogenesis

Katy Warr, Jonathon Hare, David Thomas

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The creation of future low-power neuromorphic solutions requires specialist spiking neural network (SNN) algorithms that are optimized for neuromorphic settings. One such algorithmic challenge is the ability to recall learned patterns from their noisy variants. Solutions to this problem may be required to memorize vast numbers of patterns based on limited training data and subsequently recall the patterns in the presence of noise. To solve this problem, previous work has explored sparse associative memory (SAM)-associative memory neural models that exploit the principle of sparse neural coding observed in the brain. Research into a subcategory of SAM has been inspired by the biological process of adult neurogenesis, whereby new neurons are generated to facilitate adaptive and effective lifelong learning. Although these neurogenesis models have been demonstrated in previous research, they have limitations in terms of recall memory capacity and robustness to noise. In this article, we provide a unifying framework for characterizing a type of SAM network that has been pretrained using a learning strategy that incorporated a simple neurogenesis model. Using this characterization, we formally define network topology and threshold optimization methods to empirically demonstrate greater than 104 times improvement in memory capacity compared to previous work. We show that these optimizations can facilitate the development of networks that have reduced interneuron connectivity while maintaining high recall efficacy. This paves the way for ongoing research into fast, effective, low-power realizations of associative memory on neuromorphic platforms.

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12. Toward a Free-Response Paradigm of Decision Making in Spiking Neural Networks

Zhichao Zhu, Yang Qi, Wenlian Lu, Zhigang Wang, Lu Cao, Jianfeng Feng

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Spiking neural networks (SNNs) have attracted significant interest in the development of brain-inspired computing systems due to their energy efficiency and similarities to biological information processing. In

contrast to continuous-valued artificial neural networks, which produce results in a single step, SNNs require multiple steps during inference to achieve a desired accuracy level, resulting in a burden in real-time response and energy efficiency. Inspired by the tradeoff between speed and accuracy in human and animal decision-making processes, which exhibit correlations among reaction times, task complexity, and decision confidence, an inquiry emerges regarding how an SNN model can benefit by implementing these attributes. Here, we introduce a theory of decision making in SNNs by untangling the interplay between signal and noise. Under this theory, we introduce a new learning objective that trains an SNN not only to make the correct decisions but also to shape its confidence. Numerical experiments demonstrate that SNNs trained in this way exhibit improved confidence expression, reduced trial-to-trial variability, and shorter latency to reach the desired accuracy. We then introduce a stopping policy that can stop inference in a way that further enhances the time efficiency of SNNs. The stopping time can serve as an indicator to whether a decision is correct, akin to the reaction time in animal behavior experiments. By integrating stochasticity into decision making, this study opens up new possibilities to explore the capabilities of SNNs and advance SNNs and their applications in complex decision-making scenarios where model performance is limited.

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13. Gradual Domain Adaptation via Normalizing Flows

Shogo Sagawa, Hideitsu Hino

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Standard domain adaptation methods do not work well when a large gap exists between the source and target domains. Gradual domain adaptation is one of the approaches used to address the problem. It involves leveraging the intermediate domain, which gradually shifts from the source domain to the target domain. In previous work, it is assumed that the number of intermediate domains is large and the distance between adjacent domains is small; hence, the gradual domain adaptation algorithm, involving self-training with unlabeled data sets, is applicable. In practice, however, gradual self-training will fail because the number of intermediate domains is limited and the distance between adjacent domains is large. We propose the use of normalizing flows to deal with this problem while maintaining the framework of unsupervised domain adaptation. The proposed method learns a transformation from the distribution of the target domains to the gaussian mixture distribution via the source domain. We evaluate our proposed method by experiments using real-world data sets and confirm that it mitigates the problem we have explained and improves the classification performance.

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14. Uncovering Dynamical Equations of Stochastic Decision Models Using Data-Driven SINDy Algorithm

Brendan Lenfesty, Saugat Bhattacharyya, KongFatt Wong■Lin

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Decision formation in perceptual decision making involves sensory evidence accumulation instantiated by the temporal integration of an internal decision variable toward some decision criterion or threshold, as described by sequential sampling theoretical models. The decision variable can be represented in the form of experimentally observable neural activities. Hence, elucidating the appropriate theoretical model becomes crucial to understanding the mechanisms underlying perceptual decision formation. Existing computational methods are limited to either fitting of choice behavioral data or linear model estimation from neural activity

data. In this work, we made use of sparse identification of nonlinear dynamics (SINDy), a data-driven approach, to elucidate the deterministic linear and nonlinear components of often-used stochastic decision models within reaction time task paradigms. Based on the simulated decision variable activities of the models and assuming the noise coefficient term is known beforehand, SINDy, enhanced with approaches using multiple trials, could readily estimate the deterministic terms in the dynamical equations, choice accuracy, and decision time of the models across a range of signal-to-noise ratio values. In particular, SINDy performed the best using the more memory-intensive multi-trial approach while trial-averaging of parameters performed more moderately. The single-trial approach, although expectedly not performing as well, may be useful for real-time modeling. Taken together, our work offers alternative approaches for SINDy to uncover the dynamics in perceptual decision making and, more generally, for first-passage time problems.

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15. Context-Sensitive Processing in a Model Neocortical Pyramidal Cell With Two Sites of Input Integration

Bruce Graham, Jim Kay, William A. Phillips

https://doi.org/10.1162/neco_a_01739 | pp. 588-634

Neocortical layer 5 thick-tufted pyramidal cells are prone to exhibiting burst firing on receipt of coincident basal and apical dendritic inputs. These inputs carry different information, with basal inputs coming from feedforward sensory pathways and apical inputs coming from diverse sources that provide context in the cortical hierarchy. We explore the information processing possibilities of this burst firing using computer simulations of a noisy compartmental cell model. Simulated data on stochastic burst firing due to brief, simultaneously injected basal and apical currents allow estimation of burst firing probability for different stimulus current amplitudes. Information-theory-based partial information decomposition (PID) is used to quantify the contributions of the apical and basal input streams to the information in the cell output bursting probability. Four different operating regimes are apparent, depending on the relative strengths of the input streams, with output burst probability carrying more or less information that is uniquely contributed by either the basal or apical input, or shared and synergistic information due to the combined streams. We derive and fit transfer functions for these different regimes that describe burst probability over the different ranges of basal and apical input amplitudes. The operating regimes can be classified into distinct modes of information processing, depending on the contribution of apical input to output bursting: apical cooperation, in which both basal and apical inputs are required to generate a burst; apical amplification, in which basal input alone can generate a burst but the burst probability is modulated by apical input; apical drive, in which apical input alone can produce a burst; and apical integration, in which strong apical or basal inputs alone, as well as their combination, can generate bursting. In particular, PID and the transfer function clarify that the apical amplification mode has the features required for contextually modulated information processing.

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16. Spiking Neuron-Astrocyte Networks for Image Recognition

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From biological and artificial network perspectives, researchers have started acknowledging astrocytes as computational units mediating neural processes. Here, we propose a novel biologically inspired

neuron-astrocyte network model for image recognition, one of the first attempts at implementing astrocytes in spiking neuron networks (SNNs) using a standard data set. The architecture for image recognition has three primary units: the preprocessing unit for converting the image pixels into spiking patterns, the neuron-astrocyte network forming bipartite (neural connections) and tripartite synapses (neural and astrocytic connections), and the classifier unit. In the astrocyte-mediated SNNs, an astrocyte integrates neural signals following the simplified Postnov model. It then modulates the integrate-and-fire (IF) neurons via gliotransmission, thereby strengthening the synaptic connections of the neurons within the astrocytic territory. We develop an architecture derived from a baseline SNN model for unsupervised digit classification. The spiking neuron-astrocyte networks (SNANs) display better network performance with an optimal variance-bias trade-off than SNN alone. We demonstrate that astrocytes promote faster learning, support memory formation and recognition, and provide a simplified network architecture. Our proposed SNAN can serve as a benchmark for future researchers on astrocyte implementation in artificial networks, particularly in neuromorphic systems, for its simplified design.

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17. Active Inference and Intentional Behavior

Karl Friston, Tommaso Salvatori, Takuya Isomura, Alexander Tschantz, Alex Kiefer, Tim Verbelen, Magnus Koudahl, Aswin Paul, Thomas Parr, Adeel Razi, Brett J. Kagan, Christopher L. Buckley, Maxwell J. D. Ramstead

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Recent advances in theoretical biology suggest that key definitions of basal cognition and sentient behavior may arise as emergent properties of in vitro cell cultures and neuronal networks. Such neuronal networks reorganize activity to demonstrate structured behaviors when embodied in structured information landscapes. In this article, we characterize this kind of self-organization through the lens of the free energy principle, that is, as self-evidencing. We do this by first discussing the definitions of reactive and sentient behavior in the setting of active inference, which describes the behavior of agents that model the consequences of their actions. We then introduce a formal account of intentional behavior that describes agents as driven by a preferred end point or goal in latent state-spaces. We then investigate these forms of (reactive, sentient, and intentional) behavior using simulations. First, we simulate the in vitro experiments, in which neuronal cultures modulated activity to improve gameplay in a simplified version of Pong by implementing nested, free energy minimizing processes. The simulations are then used to deconstruct the ensuing predictive behavior, leading to the distinction between merely reactive, sentient, and intentional behavior with the latter formalized in terms of inductive inference. This distinction is further studied using simple machine learning benchmarks (navigation in a grid world and the Tower of Hanoi problem) that show how quickly and efficiently adaptive behavior emerges under an inductive form of active inference.

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18. Learning in Wilson-Cowan Model for Metapopulation

Raffaele Marino, Lorenzo Buffoni, Lorenzo Chicchi, Francesca Di Patti, Diego Febbe, Lorenzo Giambagli, Duccio Fanelli

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The Wilson-Cowan model for metapopulation, a neural mass network model, treats different subcortical regions of the brain as connected nodes, with connections representing various types of structural,

functional, or effective neuronal connectivity between these regions. Each region comprises interacting populations of excitatory and inhibitory cells, consistent with the standard Wilson-Cowan model. In this article, we show how to incorporate stable attractors into such a metapopulation model's dynamics. By doing so, we transform the neural mass network model into a biologically inspired learning algorithm capable of solving different classification tasks. We test it on MNIST and Fashion MNIST in combination with convolutional neural networks, as well as on CIFAR-10 and TF-FLOWERS, and in combination with a transformer architecture (BERT) on IMDB, consistently achieving high classification accuracy.

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19. Knowledge as a Breaking of Ergodicity

Yang He, Vassiliy Lubchenko

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We construct a thermodynamic potential that can guide training of a generative model defined on a set of binary degrees of freedom. We argue that upon reduction in description, so as to make the generative model computationally manageable, the potential develops multiple minima. This is mirrored by the emergence of multiple minima in the free energy proper of the generative model itself. The variety of training samples that employ N binary degrees of freedom is ordinarily much lower than the size 2N of the full phase space. The nonrepresented configurations, we argue, should be thought of as comprising a high-temperature phase separated by an extensive energy gap from the configurations composing the training set. Thus, training amounts to sampling a free energy surface in the form of a library of distinct bound states, each of which breaks ergodicity. The ergodicity breaking prevents escape into the near continuum of states comprising the high-temperature phase; thus, it is necessary for proper functionality. It may, however, have the side effect of limiting access to patterns that were underrepresented in the training set. At the same time, the ergodicity breaking within the library complicates both learning and retrieval. As a remedy, one may concurrently employ multiple generative models-up to one model per free energy minimum.

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20. Enhanced EEG Forecasting: A Probabilistic Deep Learning Approach

Hanna Pankka, Jaakko Lehtinen, Risto J. Ilmoniemi, Timo Roine

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Forecasting electroencephalography (EEG) signals, that is, estimating future values of the time series based on the past ones, is essential in many real-time EEG-based applications, such as brain-computer interfaces and closed-loop brain stimulation. As these applications are becoming more and more common, the importance of a good prediction model has increased. Previously, the autoregressive model (AR) has been employed for this task; however, its prediction accuracy tends to fade quickly as multiple steps are predicted. We aim to improve on this by applying probabilistic deep learning to make robust longer-range forecasts. For this, we applied the probabilistic deep neural network model WaveNet to forecast resting-state EEG in theta- (4-7.5 Hz) and alpha-frequency (8-13 Hz) bands and compared it to the AR model. WaveNet reliably predicted EEG signals in both theta and alpha frequencies 150 ms ahead, with mean absolute errors of 1.0 ± 1.1 μV (theta) and 0.9 ± 1.1 μV (alpha), and outperformed the AR model in estimating the signal amplitude and phase. Furthermore, we found that the probabilistic approach offers a way of forecasting even more accurately while effectively discarding uncertain predictions. We demonstrate for the first time that probabilistic deep learning can be used to forecast resting-state EEG time series. In the

future, the developed model can enhance the real-time estimation of brain states in brain-computer interfaces and brain stimulation protocols. It may also be useful for answering neuroscientific questions and for diagnostic purposes.

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21. Nearly Optimal Learning Using Sparse Deep ReLU Networks in Regularized Empirical Risk Minimization With Lipschitz Loss

Ke Huang, Mingming Liu, Shujie Ma

https://doi.org/10.1162/neco_a_01742 | pp. 815-870

We propose a sparse deep ReLU network (SDRN) estimator of the regression function obtained from regularized empirical risk minimization with a Lipschitz loss function. Our framework can be applied to a variety of regression and classification problems. We establish novel nonasymptotic excess risk bounds for our SDRN estimator when the regression function belongs to a Sobolev space with mixed derivatives. We obtain a new, nearly optimal, risk rate in the sense that the SDRN estimator can achieve nearly the same optimal minimax convergence rate as one-dimensional nonparametric regression with the dimension involved in a logarithm term only when the feature dimension is fixed. The estimator has a slightly slower rate when the dimension grows with the sample size. We show that the depth of the SDRN estimator grows with the sample size in logarithmic order, and the total number of nodes and weights grows in polynomial order of the sample size to have the nearly optimal risk rate. The proposed SDRN can go deeper with fewer parameters to well estimate the regression and overcome the overfitting problem encountered by conventional feedforward neural networks.

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22. A Generalized Time Rescaling Theorem for Temporal Point Processes

Xi Zhang, Akshay Aravamudan, Georgios C. Anagnostopoulos

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Temporal point processes are essential for modeling event dynamics in fields such as neuroscience and social media. The time rescaling theorem is commonly used to assess model fit by transforming a point process into a homogeneous Poisson process. However, this approach requires that the process be nonterminating and that complete (hence, unbounded) realizations are observed-conditions that are often unmet in practice. This article introduces a generalized time-rescaling theorem to address these limitations and, as such, facilitates a more widely applicable evaluation framework for point process models in diverse real-world scenarios.

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23. Elucidating the Theoretical Underpinnings of Surrogate Gradient Learning in Spiking Neural Networks

Julia Gygax, Friedemann Zenke

https://doi.org/10.1162/neco_a_01752 | pp. 886-925

Training spiking neural networks to approximate universal functions is essential for studying information processing in the brain and for neuromorphic computing. Yet the binary nature of spikes poses a challenge for direct gradient-based training. Surrogate gradients have been empirically successful in circumventing

this problem, but their theoretical foundation remains elusive. Here, we investigate the relation of surrogate gradients to two theoretically well-founded approaches. On the one hand, we consider smoothed probabilistic models, which, due to the lack of support for automatic differentiation, are impractical for training multilayer spiking neural networks but provide derivatives equivalent to surrogate gradients for single neurons. On the other hand, we investigate stochastic automatic differentiation, which is compatible with discrete randomness but has not yet been used to train spiking neural networks. We find that the latter gives surrogate gradients a theoretical basis in stochastic spiking neural networks, where the surrogate derivative matches the derivative of the neuronal escape noise function. This finding supports the effectiveness of surrogate gradients in practice and suggests their suitability for stochastic spiking neural networks. However, surrogate gradients are generally not gradients of a surrogate loss despite their relation to stochastic automatic differentiation. Nevertheless, we empirically confirm the effectiveness of surrogate gradients in stochastic multilayer spiking neural networks and discuss their relation to deterministic networks as a special case. Our work gives theoretical support to surrogate gradients and the choice of a suitable surrogate derivative in stochastic spiking neural networks.

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24. The Leaky Integrate-and-Fire Neuron Is a Change-Point Detector for Compound Poisson Processes

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Animal nervous systems can detect changes in their environments within hundredths of a second. They do so by discerning abrupt shifts in sensory neural activity. Many neuroscience studies have employed change-point detection (CPD) algorithms to estimate such abrupt shifts in neural activity. But very few studies have suggested that spiking neurons themselves are online change-point detectors. We show that a leaky integrate-and-fire (LIF) neuron implements an online CPD algorithm for a compound Poisson process. We quantify the CPD performance of an LIF neuron under various regions of its parameter space. We show that CPD can be a recursive algorithm where the output of one algorithm can be input to another. Then we show that a simple feedforward network of LIF neurons can quickly and reliably detect very small changes in input spiking rates. For example, our network detects a 5% change in input rates within 20 ms on average, and false-positive detections are extremely rare. In a rigorous statistical context, we interpret the salient features of the LIF neuron: its membrane potential, synaptic weight, time constant, resting potential, action potentials, and threshold. Our results potentially generalize beyond the LIF neuron model and its associated CPD problem. If spiking neurons perform change-point detection on their inputs, then the electrophysiological properties of their membranes must be related to the spiking statistics of their inputs. We demonstrate one example of this relationship for the LIF neuron and compound Poisson processes and suggest how to test this hypothesis more broadly. Maybe neurons are not noisy devices whose action potentials must be averaged over time or populations. Instead, neurons might implement sophisticated, optimal, and online statistical algorithms on their inputs.

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25. Adding Space to Random Networks of Spiking Neurons: A Method Based on Scaling the Network Size

Cecilia Romaro, José Roberto Castilho Piqueira, Antônio C. Roque

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Many spiking neural network models are based on random graphs that do not include topological and structural properties featured in real brain networks. To turn these models into spatial networks that describe the topographic arrangement of connections is a challenging task because one has to deal with neurons at the spatial network boundary. Addition of space may generate spurious network behavior like oscillations introduced by periodic boundary conditions or unbalanced neuronal spiking due to lack or excess of connections. Here, we introduce a boundary solution method for networks with added spatial extension that prevents the occurrence of spurious spiking behavior. The method is based on a recently proposed technique for scaling the network size that preserves first- and second-order statistics.

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26. Distributed Synaptic Connection Strength Changes Dynamics in a Population Firing Rate Model in Response to Continuous External Stimuli

Masato Sugino, Mai Tanaka, Kenta Shimba, Kiyoshi Kotani, Yasuhiko Jimbo

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Neural network complexity allows for diverse neuronal population dynamics and realizes higherorder brain functions such as cognition and memory. Complexity is enhanced through chemical synapses with exponentially decaying conductance and greater variation in the neuronal connection strength due to synaptic plasticity. However, in the macroscopic neuronal population model, synaptic connections are often described by spike connections, and connection strengths within the population are assumed to be uniform. Thus, the effects of synaptic connections variation on network synchronization remain unclear. Based on recent advances in mean field theory for the quadratic integrate-and-fire neuronal network model, we introduce synaptic conductance and variation of connection strength into the excitatory and inhibitory neuronal population model and derive the macroscopic firing rate equations for faithful modeling. We then introduce a heuristic switching rule of the dynamic system with respect to the mean membrane potentials to avoid divergences in the computation caused by variations in the neuronal connection strength. We show that the switching rule agrees with the numerical computation of the microscopic level model. In the derived model, variations in synaptic conductance and connection strength strongly alter the stability of the solutions to the equations, which is related to the mechanism of synchronous firing. When we apply physiologically plausible values from layer 4 of the mammalian primary visual cortex to the derived model, we observe event-related desynchronization at the alpha and beta frequencies and event-related synchronization at the gamma frequency over a wide range of balanced external currents. Our results show that the introduction of complex synaptic connections and physiologically valid numerical values into the low-dimensional mean field equations reproduces dynamic changes such as eventrelated (de)synchronization, and provides a unique mathematical insight into the relationship between synaptic strength variation and oscillatory mechanism.

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27. Multilevel Data Representation for Training Deep Helmholtz Machines

Jose Miguel Ramos, Luis Sa-Couto, Andreas Wichert

https://doi.org/10.1162/neco_a_01748 | pp. 1010-1033

A vast majority of the current research in the field of machine learning is done using algorithms with strong arguments pointing to their biological implausibility such as backpropagation, deviating the field's focus from understanding its original organic inspiration to a compulsive search for optimal performance. Yet there have been a few proposed models that respect most of the biological constraints present in the human brain and are valid candidates for mimicking some of its properties and mechanisms. In this letter, we focus on guiding the learning of a biologically plausible generative model called the Helmholtz machine in complex search spaces using a heuristic based on the human image perception mechanism. We hypothesize that this model's learning algorithm is not fit for deep networks due to its Hebbian-like local update rule, rendering it incapable of taking full advantage of the compositional properties that multilayer networks provide. We propose to overcome this problem by providing the network's hidden layers with visual queues at different resolutions using multilevel data representation. The results on several image data sets showed that the model was able to not only obtain better overall quality but also a wider diversity in the generated images, corroborating our intuition that using our proposed heuristic allows the model to take more advantage of the network's depth growth. More important, they show the unexplored possibilities underlying brain-inspired models and techniques.

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28. Reformulation of RBM to Unify Linear and Nonlinear Dimensionality Reduction

Jiangsheng You, Chun■Yen Liu

https://doi.org/10.1162/neco_a_01751 | pp. 1034-1055

A restricted Boltzmann machine (RBM) is a two-layer neural network with shared weights and has been extensively studied for dimensionality reduction, data representation, and recommendation systems in the literature. The traditional RBM requires a probabilistic interpretation of the values on both layers and a Markov chain Monte Carlo (MCMC) procedure to generate samples during the training. The contrastive divergence (CD) is efficient to train the RBM, but its convergence has not been proved mathematically. In this letter, we investigate the RBM by using a maximum a posteriori (MAP) estimate and the expectation-maximization (EM) algorithm. We show that the CD algorithm without MCMC is convergent for the conditional likelihood object function. Another key contribution in this letter is the reformulation of the RBM into a deterministic model. Within the reformulated RBM, the CD algorithm without MCMC approximates the gradient descent (GD) method. This reformulated RBM can take the continuous scalar and vector variables on the nodes with flexibility in choosing the activation functions. Numerical experiments show its capability in both linear and nonlinear dimensionality reduction, and for the nonlinear dimensionality reduction, the reformulated RBM can outperform principal component analysis (PCA) by choosing the proper activation functions. Finally, we demonstrate its application to vector-valued nodes for the CIFAR-10 data set (color images) and the multivariate sequence data, which cannot be configured naturally with the traditional RBM. This work not only provides theoretical insights regarding the traditional RBM but also unifies the linear and nonlinear dimensionality reduction for scalar and vector variables.

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29. Dynamics of Continuous Attractor Neural Networks With Spike Frequency Adaptation

Yujun Li, Tianhao Chu, Si Wu

https://doi.org/10.1162/neco_a_01757 | pp. 1057-1101

Attractor neural networks consider that neural information is stored as stationary states of a dynamical system formed by a large number of interconnected neurons. The attractor property empowers a neural system to encode information robustly, but it also incurs the difficulty of rapid update of network states, which can impair information update and search in the brain. To overcome this difficulty, a solution is to include adaptation in the attractor network dynamics, whereby the adaptation serves as a slow negative feedback mechanism to destabilize what are otherwise permanently stable states. In such a way, the neural system can, on one hand, represent information reliably using attractor states, and on the other hand, perform computations wherever rapid state updating is involved. Previous studies have shown that continuous attractor neural networks with adaptation (A-CANNs) exhibit rich dynamical behaviors accounting for various brain functions. In this review, we present a comprehensive view of the rich diverse dynamics of A-CANNs. Moreover, we provide a unified mathematical framework to understand these different dynamical behaviors and briefly discuss their biological implications.

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30. Dynamics and Bifurcation Structure of a Mean-Field Model of Adaptive Exponential Integrate-and-Fire Networks

Lionel Kusch, Damien Depannemaecker, Alain Destexhe, Viktor Jirsa

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The study of brain activity spans diverse scales and levels of description and requires the development of computational models alongside experimental investigations to explore integrations across scales. The high dimensionality of spiking networks presents challenges for understanding their dynamics. To tackle this, a mean-field formulation offers a potential approach for dimensionality reduction while retaining essential elements. Here, we focus on a previously developed mean-field model of adaptive exponential integrate and fire (AdEx) networks used in various research work. We observe qualitative similarities in the bifurcation structure but quantitative differences in mean firing rates between the mean-field model and AdEx spiking network simulations. Even if the mean-field model does not accurately predict phase shift during transients and oscillatory input, it generally captures the qualitative dynamics of the spiking network's response to both constant and varying inputs. Finally, we offer an overview of the dynamical properties of the AdExMF to assist future users in interpreting their results of simulations.

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31. Neural Code Translation With LIF Neuron Microcircuits

Ville Karlsson, Joni Kämäräinen

https://doi.org/10.1162/neco_a_01754 | pp. 1124-1153

Spiking neural networks (SNNs) provide an energy-efficient alternative to traditional artificial neural networks, leveraging diverse neural encoding schemes such as rate, time-to-first-spike (TTFS), and population-based binary codes. Each encoding method offers distinct advantages: TTFS enables rapid and precise transmission with minimal energy use, rate encoding provides robust signal representation, and binary population encoding aligns well with digital hardware implementations. This letter introduces a set of neural microcircuits based on leaky integrate-and-fire neurons that enable translation between these encoding schemes. We propose two applications showcasing the utility of these microcircuits. First, we

demonstrate a number comparison operation that significantly reduces spike transmission by switching from rate to TTFS encoding. Second, we present a high-bandwidth neural transmitter capable of encoding and transmitting binary population-encoded data through a single axon and reconstructing it at the target site. Additionally, we conduct a detailed analysis of these microcircuits, providing quantitative metrics to assess their efficiency in terms of neuron count, synaptic complexity, spike overhead, and runtime. Our findings highlight the potential of LIF neuron microcircuits in computational neuroscience and neuromorphic computing, offering a pathway to more interpretable and efficient SNN designs.

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32. Memory States From Almost Nothing: Representing and Computing in a Nonassociative Algebra

Stefan Reimann

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This letter presents a nonassociative algebraic framework for the representation and computation of information items in high-dimensional space. This framework is consistent with the principles of spatial computing and with the empirical findings in cognitive science about memory. Computations are performed through a process of multiplication-like binding and nonassociative interference-like bundling. Models that rely on associative bundling typically lose order information, which necessitates the use of auxiliary order structures, such as position markers, to represent sequential information that is important for cognitive tasks. In contrast, the nonassociative bundling proposed allows the construction of sparse representations of arbitrarily long sequences that maintain their temporal structure across arbitrary lengths. In this operation, noise is a constituent element of the representation of order information rather than a means of obscuring it. The nonassociative nature of the proposed framework results in the representation of a single sequence by two distinct states. The L-state, generated through left-associative bundling, continuously updates and emphasizes a recency effect, while the R-state, formed through right-associative bundling, encodes finite sequences or chunks, capturing a primacy effect. The construction of these states may be associated with activity in the prefrontal cortex in relation to short-term memory and hippocampal encoding in long-term memory, respectively. The accuracy of retrieval is contingent on a decision-making process that is based on the mutual information between the memory states and the cue. The model is able to replicate the serial position curve, which reflects the empirical recency and primacy effects observed in cognitive experiments.

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33. Low-Rank, High-Order Tensor Completion via t- Product-Induced Tucker (tTucker) Decomposition

Yaodong Li, Jun Tan, Peilin Yang, Guoxu Zhou, Qibin Zhao

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Recently, tensor singular value decomposition (t-SVD)-based methods were proposed to solve the low-rank tensor completion (LRTC) problem, which has achieved unprecedented success on image and video inpainting tasks. The t-SVD is limited to process third-order tensors. When faced with higher-order tensors, it reshapes them into third-order tensors, leading to the destruction of interdimensional correlations. To address this limitation, this letter introduces a tproductinduced Tucker decomposition (tTucker) model that replaces the mode product in Tucker decomposition with t-product, which jointly extends the ideas of t-SVD and high-order SVD. This letter defines the rank of the tTucker decomposition and presents an LRTC model

that minimizes the induced Schatten-p norm. An efficient alternating direction multiplier method (ADMM) algorithm is developed to optimize the proposed LRTC model, and its effectiveness is demonstrated through experiments conducted on both synthetic and real data sets, showcasing excellent performance.

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34. A Survey on Artificial Neural Networks in Human—Robot Interaction

Aleksandra ■wietlicka

https://doi.org/10.1162/neco_a_01764 | pp. 1193-1255

Artificial neural networks (ANNs) have shown great potential in enhancing human-robot interaction (HRI). ANNs are computational models inspired by the structure and function of biological neural networks in the brain, which can learn from examples and generalize to new situations. ANNs can be used to enable robots to interact with humans in a more natural and intuitive way by allowing them to recognize human gestures and expressions, understand natural language, and adapt to the environment. ANNs can also be used to improve robot autonomy, allowing robots to learn from their interactions with humans and to make more informed decisions. However, there are also challenges to using ANNs in HRI, including the need for large amounts of training data, issues with explainability, and the potential for bias. This review explores the current state of research on ANNs in HRI, highlighting both the opportunities and challenges of this approach and discussing potential directions for future research. The AI contribution involves applying ANNs to various aspects of HRI, while the application in engineering involves using ANNs to develop more interactive and intuitive robotic systems.

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35. Decision Threshold Learning in the Basal Ganglia for Multiple Alternatives

Thom Griffith, Sophie-Anne Baker, Nathan F. Lepora

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In recent years, researchers have integrated the historically separate, reinforcement learning (RL), and evidence-accumulation-to-bound approaches to decision modeling. A particular outcome of these efforts has been the RL-DDM, a model that combines value learning through reinforcement with a diffusion decision model (DDM). While the RL-DDM is a conceptually elegant extension of the original DDM, it faces a similar problem to the DDM in that it does not scale well to decisions with more than two options. Furthermore, in its current form, the RL-DDM lacks flexibility when it comes to adapting to rapid, context-cued changes in the reward environment. The question of how to best extend combined RL and DDM models so they can handle multiple choices remains open. Moreover, it is currently unclear how these algorithmic solutions should map to neurophysical processes in the brain, particularly in relation to so-called go/no-go-type models of decision making in the basal ganglia. Here, we propose a solution that addresses these issues by combining a previously proposed decision model based on the multichoice sequential probability ratio test (MSPRT), with a dual-pathway model of decision threshold learning in the basal ganglia region of the brain. Our model learns decision thresholds to optimize the trade-off between time cost and the cost of errors and so efficiently allocates the amount of time for decision deliberation. In addition, the model is context dependent and hence flexible to changes to the speed-accuracy trade-off (SAT) in the environment. Furthermore, the model reproduces the magnitude effect, a phenomenon seen experimentally in value-based decisions and is agnostic to the types of evidence and so can be used on perceptual decisions, value-based decisions, and

other types of modeled evidence. The broader significance of the model is that it contributes to the active research area of how learning systems interact by linking the previously separate models of RL-DDM to dopaminergic models of motivation and risk taking in the basal ganglia, as well as scaling to multiple alternatives.

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36. Rapid Memory Encoding in a Spiking Hippocampus Circuit Model

Jiashuo Wang, Mengwen Yuan, Jiangrong Shen, Qingao Chai, Huajin Tang

https://doi.org/10.1162/neco_a_01762 | pp. 1320-1352

Memory is a complex process in the brain that involves the encoding, consolidation, and retrieval of previously experienced stimuli. The brain is capable of rapidly forming memories of sensory input. However, applying the memory system to real-world data poses challenges in practical implementation. This article demonstrates that through the integration of sparse spike pattern encoding scheme population tempotron, and various spike-timing-dependent plasticity (STDP) learning rules, supported by bounded weights and biological mechanisms, it is possible to rapidly form stable neural assemblies of external sensory inputs in a spiking neural circuit model inspired by the hippocampal structure. The model employs neural ensemble module and competitive learning strategies that mimic the pattern separation mechanism of the hippocampal dentate gyrus (DG) area to achieve nonoverlapping sparse coding. It also uses population tempotron and NMDA-(N-methyl-D-aspartate)mediated STDP to construct associative and episodic memories, analogous to the CA3 and CA1 regions. These memories are represented by strongly connected neural assemblies formed within just a few trials. Overall, this model offers a robust computational framework to accommodate rapid memory throughout the brain-wide memory process.

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37. Excitation–Inhibition Balance Controls Synchronization in a Simple Model of Coupled Phase Oscillators

Satoshi Kuroki, Kenji Mizuseki

https://doi.org/10.1162/neco_a_01763 | pp. 1353-1372

Collective neuronal activity in the brain synchronizes during rest and desynchronizes during active behaviors, influencing cognitive processes such as memory consolidation, knowledge abstraction, and creative thinking. These states involve significant modulation of inhibition, which alters the excitation-inhibition (EI) balance of synaptic inputs. However, the influence of the EI balance on collective neuronal oscillation remains only partially understood. In this study, we introduce the EI-Kuramoto model, a modified version of the Kuramoto model, in which oscillators are categorized into excitatory and inhibitory groups with four distinct interaction types: excitatory-excitatory, excitatory-inhibitory, inhibitory-excitatory, and inhibitory-inhibitory. Numerical simulations identify three dynamic states-synchronized, bistable, and desynchronized-that can be controlled by adjusting the strength of the four interaction types. Theoretical analysis further demonstrates that the balance among these interactions plays a critical role in determining the dynamic states. This study provides valuable insights into the role of EI balance in synchronizing coupled oscillators and neurons.

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38. Predictive Coding Model Detects Novelty on Different Levels of Representation Hierarchy

Tianjin Li, Mufeng Tang, Rafa■ Bogacz

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Novelty detection, also known as familiarity discrimination or recognition memory, refers to the ability to distinguish whether a stimulus has been seen before. It has been hypothesized that novelty detection can naturally arise within networks that store memory or learn efficient neural representation because these networks already store information on familiar stimuli. However, existing computational models supporting this idea have yet to reproduce the high capacity of human recognition memory, leaving the hypothesis in question. This article demonstrates that predictive coding, an established model previously shown to effectively support representation learning and memory, can also naturally discriminate novelty with high capacity. The predictive coding model includes neurons encoding prediction errors, and we show that these neurons produce higher activity for novel stimuli, so that the novelty can be decoded from their activity. Additionally, hierarchical predictive coding networks detect novelty at different levels of abstraction within the hierarchy, from low-level sensory features like arrangements of pixels to high-level semantic features like object identities. Overall, based on predictive coding, this article establishes a unified framework that brings together novelty detection, associative memory, and representation learning, demonstrating that a single model can capture these various cognitive functions.

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39. A Categorical Framework for Quantifying Emergent Effects in Network Topology

Johnny Jingze Li, S. Guerra, Kalyan Basu, Gabriel A. Silva

https://doi.org/10.1162/neco_a_01766 | pp. 1409-1438

Emergent effect is crucial to understanding the properties of complex systems that do not appear in their basic units, but there has been a lack of theories to measure and understand its mechanisms. In this letter, we consider emergence as a kind of structural nonlinearity, discuss a framework based on homological algebra that encodes emergence as the mathematical structure of cohomologies, and then apply it to network models to develop a computational measure of emergence. This framework ties the potential for emergent effects of a system to its network topology and local structures, paving the way to predict and understand the cause of emergent effects. We show in our numerical experiment that our measure of emergence correlates with the existing information-theoretic measure of emergence.

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40. Continuous-Time Neural Networks Can Stably Memorize Random Spike Trains

Hugo Aguettaz, Hans■Andrea Loeliger

https://doi.org/10.1162/neco_a_01768 | pp. 1439-1468

This letter explores the capability of continuous-time recurrent neural networks to store and recall precisely timed scores of spike trains. We show (by numerical experiments) that this is indeed possible: within some range of parameters, any random score of spike trains (for all neurons in the network) can be robustly memorized and autonomously reproduced with stable accurate relative timing of all spikes, with probability

close to one. We also demonstrate associative recall under noisy conditions. In these experiments, the required synaptic weights are computed offline to satisfy a template that encourages temporal stability.

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41. Nonlinear Neural Dynamics and Classification Accuracy in Reservoir Computing

Claus Metzner, Achim Schilling, Andreas Maier, Patrick Krauß
https://doi.org/10.1162/neco_a_01770 | pp. 1469-1504

Reservoir computing information processing based on untrained recurrent neural networks with random connections is expected to depend on the nonlinear properties of the neurons and the resulting oscillatory, chaotic, or fixed-point dynamics of the network. However, the degree of nonlinearity required and the range of suitable dynamical regimes for a given task remain poorly understood. To clarify these issues, we study the classification accuracy of a reservoir computer in artificial tasks of varying complexity while tuning both the neuron's degree of nonlinearity and the reservoir's dynamical regime. We find that even with activation functions of extremely reduced nonlinearity, weak recurrent interactions, and small input signals, the reservoir can compute useful representations. These representations, detectable only in higher-order principal components, make complex classification tasks linearly separable for the readout layer. Increasing the recurrent coupling leads to spontaneous dynamical behavior. Nevertheless, some input-related computations can "ride on top" of oscillatory or fixed-point attractors with little loss of accuracy, whereas chaotic dynamics often reduces task performance. By tuning the system through the full range of dynamical phases, we observe in several classification tasks that accuracy peaks at both the oscillatory/chaotic and chaotic/fixed-point phase boundaries, supporting the edge of chaos hypothesis. We also present a regression task with the opposite behavior. Our findings, particularly the robust weakly nonlinear operating regime, may offer new perspectives for both technical and biological neural networks with random connectivity.

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42. Crystal-LSBO: Automated Design of De Novo Crystals With Latent Space Bayesian Optimization

Onur Boyar, Yanheng Gu, Yuji Tanaka, Shunsuke Tonogai, Tomoya Itakura, Ichiro Takeuchi
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Generative modeling of crystal structures is significantly challenged by the complexity of input data, which constrains the ability of these models to explore and discover novel crystals. This complexity often confines de novo design methodologies to merely small perturbations of known crystals and hampers the effective application of advanced optimization techniques. One such optimization technique, latent space Bayesian optimization (LSBO), has demonstrated promising results in uncovering novel objects across various domains, especially when combined with variational autoencoders (VAEs). Recognizing LSBO's potential and the critical need for innovative crystal discovery, we introduce Crystal-LSBO, a de novo design framework for crystals specifically tailored to enhance explorability within LSBO frameworks. Crystal-LSBO employs multiple VAEs, each dedicated to a distinct aspect of crystal structure-lattice, coordinates, and chemical elements-orchestrated by an integrative model that synthesizes these components into a cohesive output. This setup not only streamlines the learning process but also produces explorable latent spaces thanks to the decreased complexity of the learning task for each model, enabling LSBO approaches to

operate. Our study pioneers the use of LSBO for de novo crystal design, demonstrating its efficacy through optimization tasks focused mainly on formation energy values. Our results highlight the effectiveness of our methodology, offering a new perspective for de novo crystal discovery.

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43. Synergistic Pathways of Modulation Enable Robust Task Packing Within Neural Dynamics

Giacomo Vedovati, ShiNung Ching

https://doi.org/10.1162/neco.a.18 | pp. 1529-1550

Understanding how brain networks learn and manage multiple tasks simultaneously is of interest in both neuroscience and artificial intelligence. In this regard, a recent research thread in theoretical neuroscience has focused on how recurrent neural network models and their internal dynamics enact multi-task learning. To manage different tasks requires a mechanism to convey information about task identity or context into the model, which from a biological perspective may involve mechanisms of neuromodulation. In this study, we use recurrent network models to probe the distinctions between two forms of contextual modulation of neural dynamics, at the level of neuronal excitability and at the level of synaptic strength. We characterize these mechanisms in terms of their functional outcomes, focusing on their robustness to context ambiguity and, relatedly, their efficiency with respect to packing multiple tasks into finite size networks. We also demonstrate distinction between these mechanisms at the level of the neuronal dynamics they induce. Together, these characterizations indicate complementarity and synergy in how these mechanisms act, potentially over multiple time-scales, toward enhancing robustness of multi-task learning.

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44. Exploring the Architectural Biases of the Cortical Microcircuit

Aishwarya Balwani, Suhee Cho, Hannah Choi

https://doi.org/10.1162/neco.a.23 | pp. 1551-1599

The cortex plays a crucial role in various perceptual and cognitive functions, driven by its basic unit, the canonical cortical microcircuit. Yet, we remain short of a framework that definitively explains the structure-function relationships of this fundamental neuroanatomical motif. To better understand how physical substrates of cortical circuitry facilitate their neuronal dynamics, we employ a computational approach using recurrent neural networks and representational analyses. We examine the differences manifested by the inclusion and exclusion of biologically motivated interareal laminar connections on the computational roles of different neuronal populations in the microcircuit of hierarchically related areas throughout learning. Our findings show that the presence of feedback connections correlates with the functional modularization of cortical populations in different layers and provides the microcircuit with a natural inductive bias to differentiate expected and unexpected inputs at initialization, which we justify mathematically. Furthermore, when testing the effects of training the microcircuit and its variants with a predictive-coding-inspired strategy, we find that doing so helps better encode noisy stimuli in areas of the cortex that receive feedback, all of which combine to suggest evidence for a predictive-coding mechanism serving as an intrinsic operative logic in the cortex.

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45. Measuring Stimulus Information Transfer Between Neural Populations Through the Communication Subspace

Oren Weiss, Ruben Coen-Cagli

https://doi.org/10.1162/neco.a.17 | pp. 1600-1647

Sensory processing arises from the communication between neural populations across multiple brain areas. While the widespread presence of neural response variability shared throughout a neural population limits the amount of stimulus-related information those populations can accurately represent, how this variability affects the interareal communication of sensory information is unknown. We propose a mathematical framework to understand the impact of neural population response variability on sensory information transmission. We combine linear Fisher information, a metric connecting stimulus representation and variability, with the framework of communication subspaces, which suggests that functional mappings between cortical populations are low-dimensional relative to the space of population activity patterns. From this, we partition Fisher information depending on the alignment between the population covariance and the mean tuning direction projected onto the communication subspace or its orthogonal complement. We provide mathematical and numerical analyses of our proposed decomposition of Fisher information and examine theoretical scenarios that demonstrate how to leverage communication subspaces for flexible routing and gating of stimulus information. This work will provide researchers investigating interareal communication with a theoretical lens through which to understand sensory information transmission and guide experimental design.

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46. Toward Generalized Entropic Sparsification for Convolutional Neural Networks

Tin Barisin, Illia Horenko

https://doi.org/10.1162/neco.a.21 | pp. 1648-1676

Convolutional neural networks (CNNs) are reported to be overparametrized. The search for optimal (minimal) and sufficient architecture is an NP-hard problem: if the network has N neurons, then there are 2N possibilities to connect them-and therefore 2N possible architectures and 2N Boolean hyperparameters to encode them. Selecting the best possible hyperparameter out of them becomes an Np-hard problem since 2N grows in N faster then any polynomial Np. Here, we introduce a layer-by-layer data-driven pruning method based on the mathematical idea aiming at a computationally scalable entropic relaxation of the pruning problem. The sparse subnetwork is found from the pretrained (full) CNN using the network entropy minimization as a sparsity constraint. This allows deploying a numerically scalable algorithm with a sublinear scaling cost. The method is validated on several benchmarks (architectures): on MNIST (LeNet), resulting in sparsity of 55% to 84% and loss in accuracy of just 0.1% to 0.5%, and on CIFAR-10 (VGG-16, ResNet18), resulting in sparsity of 73% to 89% and loss in accuracy of 0.1% to 0.5%.

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47. From Function to Implementation: Exploring Degeneracy in Evolved Artificial Agents

Zhimin Hu, O■ulcan Cingiler, C. Bohm, Larissa Albantakis

https://doi.org/10.1162/neco.a.19 | pp. 1677-1708

Degeneracy-the ability of different structures to perform the same function-is a fundamental feature of biological systems, contributing to their robustness and evolvability. However, the ubiquity of degeneracy in systems generated through adaptive processes complicates our understanding of the behavioral and computational strategies they employ. In this study, we investigated degeneracy in simple computational agents, known as Markov brains, trained using an artificial evolution algorithm to solve a spatial navigation task with or without associative memory. We analyzed degeneracy at three levels: behavioral, structural, and computational, with a focus on the last. Using information-theoretical concepts, Tononi et al. (1999) proposed a functional measure of degeneracy within biological networks. Here, we extended this approach to compare degeneracy across multiple networks. Using information-theoretical tools and causal analysis, we explored the computational strategies of the evolved agents and quantified their computational degeneracy. Our findings reveal a hierarchy of degenerate solutions, from varied behaviors to diverse structures and computations. Even agents with identical evolved behaviors demonstrated different underlying structures and computations. These results underscore the pervasive nature of degeneracy in neural networks, blurring the lines between the algorithmic and implementation levels in adaptive systems, and highlight the importance of advanced analytical tools to understand their complex behaviors.

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48. Fast Multigroup Gaussian Process Factor Models

Evren Gokcen, Anna I. Jasper, Adam Kohn, Christian K. Machens, Byron M. Yu

https://doi.org/10.1162/neco.a.22 | pp. 1709-1782

Gaussian processes are now commonly used in dimensionality reduction approaches tailored to neuroscience, especially to describe changes in high-dimensional neural activity over time. As recording capabilities expand to include neuronal populations across multiple brain areas, cortical layers, and cell types, interest in extending gaussian process factor models to characterize multipopulation interactions has grown. However, the cubic runtime scaling of current methods with the length of experimental trials and the number of recorded populations (groups) precludes their application to large-scale multipopulation recordings. Here, we improve this scaling from cubic to linear in both trial length and group number. We present two approximate approaches to fitting multigroup gaussian process factor models based on inducing variables and the frequency domain. Empirically, both methods achieved orders of magnitude speed-up with minimal impact on statistical performance, in simulation and on neural recordings of hundreds of neurons across three brain areas. The frequency domain approach, in particular, consistently provided the greatest runtime benefits with the fewest trade-offs in statistical performance. We further characterize the estimation biases introduced by the frequency domain approach and demonstrate effective strategies to mitigate them. This work enables a powerful class of analysis techniques to keep pace with the growing scale of multipopulation recordings, opening new avenues for exploring brain function.

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49. Diversity Deconstrains Component Limitations in Sensorimotor Control

Yorie Nakahira, Quanying Liu, Xiyu Deng, Terrence J. Sejnowski, John C. Doyle

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Human sensorimotor control is remarkably fast and accurate at the system level despite severe speed-accuracy trade-offs at the component level. The discrepancy between the contrasting speed-accuracy trade-offs at these two levels is a paradox. Meanwhile, speed accuracy trade-offs,

heterogeneity, and layered architectures are ubiquitous in nerves, skeletons, and muscles, but they have only been studied in isolation using domain-specific models. In this article, we develop a mechanistic model for how component speed-accuracy trade-offs constrain sensorimotor control that is consistent with Fitts' law for reaching. The model suggests that diversity among components deconstrains the limitations of individual components in sensorimotor control. Such diversity-enabled sweet spots (DESSs) are ubiquitous in nature, explaining why large heterogeneities exist in the components of biological systems and how natural selection routinely evolves systems with fast and accurate responses using imperfect components.

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50. Firing Rate Models as Associative Memory: Synaptic Design for Robust Retrieval

Simone Betteti, Giacomo Baggio, Francesco Bullo, Sandro Zampieri

https://doi.org/10.1162/neco.a.28 | pp. 1807-1838

Firing rate models are dynamical systems widely used in applied and theoretical neuroscience to describe local cortical dynamics in neuronal populations. By providing a macroscopic perspective of neuronal activity, these models are essential for investigating oscillatory phenomena, chaotic behavior, and associative memory processes. Despite their widespread use, the application of firing rate models to associative memory networks has received limited mathematical exploration, and most existing studies are focused on specific models. Conversely, well-established associative memory designs, such as Hopfield networks, lack key biologically relevant features intrinsic to firing rate models, including positivity and interpretable synaptic matrices reflecting the action of long-term potentiation and long-term depression. To address this gap, we propose a general framework that ensures the emergence of rescaled memory patterns as stable equilibria in the firing rate dynamics. Furthermore, we analyze the conditions under which the memories are locally and globally asymptotically stable, providing insights into constructing biologically plausible and robust systems for associative memory retrieval.

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51. Transformer Models for Signal Processing: Scaled Dot-Product Attention Implements Constrained Filtering

Terence D. Sanger

https://doi.org/10.1162/neco.a.29 | pp. 1839-1852

The remarkable success of the transformer machine learning architecture for processing language sequences far exceeds the performance of classical signal processing methods. A unique component of transformer models is the scaled dot-product attention (SDPA) layer, which does not appear to have an analog in prior signal processing algorithms. Here, we show that SDPA operates using a novel principle that projects the current state estimate onto the space spanned by prior estimates. We show that SDPA, when used for causal recursive state estimation, implements constrained state estimation in circumstances where the constraint is unknown and may be time varying. Since constraints in high-dimensional space may represent the complex relationships that define nonlinear signals and models, this suggests that the SDPA layer and transformer models leverage constrained estimation to achieve their success. This also suggests that transformers and the SPDA layer could be a computational model for previously unexplained capabilities of human behavior.

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52. Rapid Reweighting of Sensory Inputs and Predictions in Visual Perception

William Turner, Oh■Sang Kwon, Minwoo JB Kim, Hinze Hogendoorn

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A striking perceptual phenomenon has recently been described wherein people report seeing abrupt jumps in the location of a smoothly moving object ("position resets"). Here, we show that this phenomenon can be understood within the framework of recursive Bayesian estimation as arising from transient gain changes, temporarily prioritizing sensory input over predictive beliefs. From this perspective, position resets reveal a capacity for rapid adaptive precision weighting in human visual perception and offer a possible test bed within which to study the timing and flexibility of sensory gain control.

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53. Distance-Based Logistic Matrix Factorization

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Matrix factorization is a central paradigm in matrix completion and collaborative filtering. Low-rank factorizations have been extremely successful in reconstructing and generalizing high-dimensional data in a wide variety of machine learning problems from drug-target discovery to music recommendations. Virtually all proposed matrix factorization techniques use the dot product between latent factor vectors to reconstruct the original matrix. We propose a reformulation of the widely used logistic matrix factorization in which we use the distance, rather than the dot product, to measure similarity between latent factors. We show that this measure of similarity, which can draw nonlinear decision boundaries and respect triangle inequalities between points, has more expressive power and modeling capacity. The distance-based model implemented in Euclidean and hyperbolic space outperforms previous formulations of logistic matrix factorization on three different biological test problems with disparate structure and statistics. In particular, we show that a distance-based factorization (1) generalizes better to test data, (2) achieves optimal performance at lower factor space dimension, and (3) clusters data better in the latent factor space.

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54. Sequential Learning in the Dense Associative Memory

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Sequential learning involves learning tasks in a sequence and proves challenging for most neural networks. Biological neural networks regularly succeed at the sequential learning challenge and are even capable of transferring knowledge both forward and backward between tasks. Artificial neural networks often totally fail to transfer performance between tasks and regularly suffer from degraded performance or catastrophic forgetting on previous tasks. Models of associative memory have been used to investigate the discrepancy between biological and artificial neural networks due to their biological ties and inspirations, of which the Hopfield network is the most studied model. The dense associative memory (DAM), or modern Hopfield network, generalizes the Hopfield network, allowing for greater capacities and prototype learning behaviors while still retaining the associative memory structure. We give a substantial review of the sequential learning space with particular respect to the Hopfield network and associative memories. We present the first

published benchmarks of sequential learning in the DAM using various sequential learning techniques and analyze the results of the sequential learning to demonstrate previously unseen transitions in the behavior of the DAM. This letter also discusses the departure from biological plausibility that may affect the utility of the DAM as a tool for studying biological neural networks. We present our findings, including the effectiveness of a range of state-of-the-art sequential learning methods when applied to the DAM, and use these methods to further the understanding of DAM properties and behaviors.

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55. A Chimera Model for Motion Anticipation in the Retina and the Primary Visual Cortex

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We propose a mean field model of the primary visual cortex (V1), connected to a realistic retina model, to study the impact of the retina on motion anticipation. We first consider the case where the retina does not itself provide anticipation-which is then only triggered by a cortical mechanism, the "anticipation by latency"-and unravel the effects of the retinal input amplitude, of stimulus features such as speed and contrast and of the size of cortical extensions and fiber conduction speed. Then we explore the changes in the cortical wave of anticipation when V1 is triggered by retina-driven anticipatory mechanisms: gain control and lateral inhibition by amacrine cells. Here, we show how retinal and cortical anticipation combine to provide an efficient processing where the simulated cortical response is in advance over the moving object that triggers this response, compensating the delays in visual processing.

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56. Encoding of Numerosity With Robustness to Object and Scene Identity in Biologically Inspired Object Recognition Networks

Thomas Chapalain, Bertrand Thirion, Evelyn Eger
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Number sense, the ability to rapidly estimate object quantities in a visual scene without precise counting, is a crucial cognitive capacity found in humans and many other animals. Recent studies have identified artificial neurons tuned to numbers of items in biologically inspired vision models, even before training, and proposed these artificial neural networks as candidate models for the emergence of number sense in the brain. But real-world numerosity perception requires abstraction from the properties of individual objects and their contexts, unlike the simplified dot patterns used in previous studies. Using novel synthetically generated photorealistic stimuli, we show that deep convolutional neural networks optimized for object recognition encode information on approximate numerosity across diverse objects and scene types, which could be linearly read out from distributed activity patterns of later convolutional layers of different network architectures tested. In contrast, untrained networks with random weights failed to represent numerosity with abstractness to other visual properties and instead captured mainly low-level visual features. Our findings emphasize the importance of using complex, naturalistic stimuli to investigate mechanisms of number sense in both biological and artificial systems, and they suggest that the capacity of untrained networks to account for early-life numerical abilities should be reassessed. They further point to a possible, so far underappreciated, contribution of the brain's ventral visual pathway to representing numerosity with abstractness to other high-level visual properties.

57. Modeling Higher-Order Interactions in Sparse and Heavy-Tailed Neural Population Activity

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Neurons process sensory stimuli efficiently, showing sparse yet highly variable ensemble spiking activity involving structured higher-order interactions. Notably, while neural populations are mostly silent, they occasionally exhibit highly synchronous activity, resulting in sparse and heavy-tailed spike-count distributions. However, its mechanistic origin-specifically, what types of nonlinear properties in individual neurons induce such population-level patterns-remains unclear. In this study, we derive sufficient conditions under which the joint activity of homogeneous binary neurons generates sparse and widespread population firing rate distributions in infinitely large networks. We then propose a subclass of exponential family distributions that satisfy this condition. This class incorporates structured higher-order interactions with alternating signs and shrinking magnitudes, along with a base-measure function that offsets distributional concentration, giving rise to parameter-dependent sparsity and heavy-tailed population firing rate distributions. Analysis of recurrent neural networks that recapitulate these distributions reveals that individual neurons possess threshold-like nonlinearity, followed by supralinear activation that jointly facilitates sparse and synchronous population activity. These nonlinear features resemble those in modern Hopfield networks, suggesting a connection between widespread population activity and the network's memory capacity. The theory establishes sparse and heavy-tailed distributions for binary patterns, forming a foundation for developing energy-efficient spike-based learning machines.

58. Feature Normalization Prevents Collapse of Noncontrastive Learning Dynamics

Han Bao

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Contrastive learning is a self-supervised representation learning framework where two positive views generated through data augmentation are made similar by an attraction force in a data representation space, while a repulsive force makes them far from negative examples. Noncontrastive learning, represented by BYOL and SimSiam, gets rid of negative examples and improves computational efficiency. While learned representations may collapse into a single point due to the lack of the repulsive force at first sight, Tian et al. (2021) revealed through learning dynamics analysis that the representations can avoid collapse if data augmentation is sufficiently stronger than regularization. However, their analysis does not take into account commonly used feature normalization, a normalizer before measuring the similarity of representations, and hence excessively strong regularization may still collapse the dynamics, an unnatural behavior under the presence of feature normalization. Therefore, we extend the previous theory based on the L2 loss by considering the cosine loss instead, which involves feature normalization. We show that the cosine loss induces sixth-order dynamics (while the L2 loss induces a third-order one), in which a stable equilibrium dynamically emerges even if there are only collapsed solutions with given initial parameters. Thus, we offer a new understanding that feature normalization plays an important role in robustly preventing the dynamics collapse.

59. Model Predictive Control on the Neural Manifold

Christof Fehrman, C. Daniel Meliza

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Neural manifolds are an attractive theoretical framework for characterizing the complex behaviors of neural populations. However, many of the tools for identifying these low-dimensional subspaces are correlational and provide limited insight into the underlying dynamics. The ability to precisely control the latent activity of a circuit would allow researchers to investigate the structure and function of neural manifolds. We simulate controlling the latent dynamics of a neural population using closed-loop, dynamically generated sensory inputs. Using a spiking neural network (SNN) as a model of a neural circuit, we find low-dimensional representations of both the network activity (the neural manifold) and a set of salient visual stimuli. The fields of classical and optimal control offer a range of methods to choose from for controlling dynamics on the neural manifold, which differ in performance, computational cost, and ease of implementation. Here, we focus on two commonly used control methods: proportional-integral-derivative (PID) control and model predictive control (MPC). PID is a computationally lightweight controller that is simple to implement. In contrast, MPC is a model-based, anticipatory controller with a much higher computational cost and engineering overhead. We evaluate both methods on trajectory-following tasks in latent space, under partial observability and in the presence of unknown noise. While both controllers in some cases were able to successfully control the latent dynamics on the neural manifold, MPC consistently produced more accurate control and required less hyperparameter tuning. These results demonstrate how MPC can be applied on the neural manifold using data-driven dynamics models and provide a framework to experimentally test for causal relationships between manifold dynamics and external stimuli.

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60. Estimating Phase From Observed Trajectories Using the Temporal 1-Form

Simon Wilshin, Matthew D. Kvalheim, Clayton Scott, Shai Revzen

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Oscillators are ubiquitous in nature and are usually associated with the existence of an asymptotic phase that governs the long-term dynamics of the oscillator. We show that the asymptotic phase can be estimated using a carefully chosen series expansion that directly computes the phase response curve (PRC) and provides an algorithm for estimating the coefficients of this series. Unlike previously available data-driven phase estimation methods, our algorithm can use observations that are much shorter than a cycle; has proven convergence rate bounds as a function of the properties of measurement noise and system noise; will recover phase within any forward invariant region for which sufficient data are available; recovers the PRCs that govern weak oscillator coupling; and recovers isochron curvature and recovers nonlinear features of isochron geometry. Our method may find application wherever models of oscillator dynamics need to be constructed from measured or simulated time-series.

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61. Boosting MCTS With Free Energy Minimization

Mawaba Pascal Dao, Adrian M. Peter

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Active inference, grounded in the free energy principle, provides a powerful lens for understanding how agents balance exploration and goal-directed behavior in uncertain environments. Here, we propose a new planning framework that integrates Monte Carlo tree search (MCTS) with active inference objectives to systematically reduce epistemic uncertainty while pursuing extrinsic rewards. Our key insight is that MCTS, already renowned for its search efficiency, can be naturally extended to incorporate free energy minimization by blending expected rewards with information gain. Concretely, the cross-entropy method (CEM) is used to optimize action proposals at the root node, while tree expansions leverage reward modeling alongside intrinsic exploration bonuses. This synergy allows our planner to maintain coherent estimates of value and uncertainty throughout planning, without sacrificing computational tractability. Empirically, we benchmark our planner on a diverse set of continuous control tasks, where it demonstrates performance gains over both stand-alone CEM and MCTS with random rollouts.

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62. Fusing Foveal Fixations Using Linear Retinal Transformations and Bayesian Experimental Design

Christopher K. I. Williams

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Humans (and many vertebrates) face the problem of fusing together multiple fixations of a scene in order to obtain a representation of the whole, where each fixation uses a high-resolution fovea and decreasing resolution in the periphery. In this letter, we explicitly represent the retinal transformation of a fixation as a linear downsampling of a high-resolution latent image of the scene, exploiting the known geometry. This linear transformation allows us to carry out exact inference for the latent variables in factor analysis (FA) and mixtures of FA models of the scene. This also allows us to formulate and solve the choice of where to look next as a Bayesian experimental design problem using the expected information gain criterion. Experiments on the Frey faces and MNIST data sets demonstrate the effectiveness of our models.